

HealthConnect Week 7: Testing, Refinement & Validation Record

Data Analytics Track | Prepared by Hillary Emmanuel | Fields follow Section 12 of the Week 7 brief. Rows marked PENDING need the live Power BI dashboard and are completed after those tests.

Component Tested	Testing Objective	Test / Scenario	Expected Result	Actual Result	Pass / Fail	Issue Identified	Action Taken	Retest Result	Collaborating Track	Evidence
KPI 1: prior no-show bracket	Does the dashboard KPI equal the source data?	Dashboard % vs source CSV	Exact match	43.5 / 53.5 / 59.4 / 68.8%	PASS	None	None	N/A	None (DA only)	Notebook Sec 1; data/processed/kpi1_ci_prior_no_show_bracket.csv
KPI 2: lead-time band	Does the dashboard KPI equal the source data?	Dashboard % vs source CSV	Exact match	27.8 / 33.6 / 43.2 / 53.8 / 67.7%	PASS	None	None	N/A	None (DA only)	Notebook Sec 1; data/processed/kpi2_ci_lead_time_band.csv
KPI 3: distance band	Does the dashboard KPI equal the source data?	Dashboard % vs source CSV	Exact match	46.5 / 46.5 / 48.5 / 54.1%	PASS	None	None	N/A	None (DA only)	Notebook Sec 1; data/processed/kpi3_ci_distance_band.csv
KPI 4: reminder sent	Does the dashboard KPI equal the source data?	Dashboard % vs source CSV	Exact match	47.4% / 51.4%	PASS	None	None	N/A	None (DA only)	Notebook Sec 1; data/processed/kpi4_ci_reminder_sent.csv
All 4 dashboard KPIs	Is the pipeline reproducible outside the notebook?	Independent recompute from the raw prepared CSV	Same values	Matched to 4 decimal places	PASS	None	None	N/A	None (DA only)	Notebook Sec 1
KPI 5: cancellation-t-o-no-show ratio	Does the ratio reproduce the Week 6 figures?	Recompute overall and by lead band; compare with the Week 6 KPI Validation report	Same point estimates	0.109 overall; 0.202 / 0.139 / 0.118 / 0.105 / 0.076 by lead band; all match	PASS	None in the data. The dashboard's KPI 5 chart (bottom right, no error bars) still has to be read in Power BI.	Added to the test suite	N/A	None (DA only)	Notebook Sec 1b; reports/test_evidence/kpi5_and_distance_checks_output.txt
Lead-time finding x appointment_type	Does the finding hold in every appointment type?	Segment test	Direction holds in every subgroup	46–60 day band has the highest no-show rate in all 4 types	PASS	None	None	N/A	None (DA only)	Notebook Sec 2
Lead-time finding x age_group	Does the finding hold in every age group?	Segment test	Direction holds in every subgroup	46–60 day band highest in all 6 age groups	PASS	None	None	N/A	None (DA only)	Notebook Sec 2
Distance band finding (KPI 3)	Is the 15 km cliff real?	CI overlap check on the 3 middle bands	CIs overlap (Week 6 downgrade holds)	All 3 consecutive pairs overlap	PASS (finding)	Chart shows four rising bars, which reads as a gradual trend.	Refinement designed as a mockup: group the 3 overlapping bands, isolate 15+ km, add annotation and a finding title.	PENDING: needs the live Power BI export	None (DA only)	Notebook Sec 3; outputs/dashboard/distance_chart_refinement_mockup.png
Distance vs lead time	Is distance redundant with lead time (Week 6 open question)?	Correlation; 15+ km vs under-15 km gap inside each lead band	Gap vanishes if redundant	Correlation –0.008; 15+ km higher in all 5 bands (+3.6 to +10.5 points, about +7 overall); 2 of 5 bands individually significant	PASS(not redundant)	None. Effect is modest and subgroups are small (118–259).	Keep the 15+ km threshold framing	N/A	None (DA only)	Notebook Sec 3b; reports/test_evidence/kpi5_and_distance_checks_output.txt
Distance model weight	Why does the model give distance little weight?	Logistic Regression AUC with vs without distance, 5-fold CV	Small gain if the effect is modest	0.6732 without, 0.6772 with distance (+0.004, inside ±0.01 fold noise)	PASS	None	None	N/A	DS track (model check)	Notebook Sec 3b
Double-risk segment reproduction	Does the Week 6 segment (~4.7% / ~73%) reproduce?	lead ≥46 days and prior no-shows ≥2	~4.7% of bookings, ~73% risk	2.46% / 82.1%	FAIL, then FIXED	Wrong lead-time threshold (46 instead of 31).	Corrected to ≥31. The Week 6 Decision Support text also says 46+; superseded.	4.66% / 73.0%, exact match	None (led to the cross-track test)	Notebook Sec 4

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Double-risk segment stability	Is the corrected segment stable?	Bootstrap on the corrected segment (n=233)	Reasonably tight CI	95% CI [67.4%, 79.0%], SE about 2.9 points	PASS	None	None	N/A	None (DA only)	Notebook Sec 4
DS collaborator's double_risk_flag (Random Forest)	Does the DS feature encode the DA finding it cites?	Compare coded ≥ 46 with validated ≥ 31 ; 5-fold CV for both	Same segment; no performance effect	Coded: 2.46% / 82.1%, leaves out 110 patients (62.7%). AUC 0.6661 vs 0.6662	FAIL (docs), then FIXED	Code and comment disagree: coded ≥ 46 , cited finding ≥ 31 .	Standalone retest with ≥ 31 ; decision to keep 31+; handoff package prepared.	AUC 0.6661 vs 0.6662, no change	DS track collaborator	Report Sec 6; reports/cross_track_evidence/crosstrack_double_risk_flag_retest_evidence.csv; handoff/ds_track_collaborator/
DS collaborator's double_risk_flag (exact Week 6 candidate, Logistic Regression)	Does the threshold matter for the model that cannot learn interactions alone?	Week 6 candidate rebuilt from its own code (12 features, scaled, balanced); flag ≥ 46 vs ≥ 31 vs no flag; Week 6 CV method and 5 $\times$ 5 repeated CV	Threshold could matter more	Candidate reproduced (0.6819 $\pm$ 0.0120). ≥ 46 : 0.6819; ≥ 31 : 0.6821; no flag: 0.6820. 5 $\times$ 5 CV paired difference +0.0001, 31+ wins 13/25 folds	PASS	None beyond the flag mismatch already logged. Model counts differ from the DA counts because the model excludes cancelled rows (120 and 225 on 4,737 rows).	31+ kept.	N/A	DS track collaborator	scripts/week7_double_risk_exact_candidate_test.py; reports/cross_track_evidence/ (exact-candidate CSV and log)
Dashboard filters / slicers	Do dashboard values stay correct when filtered?	Add slicers (appointment_type, age_group, reminder_sent); compare cards and charts with expected values for T0 to T3	Match dashboard_filter_expected_values.csv	T0: 5000/2423/263/48.5%; T1 (Follow-up): 1421/728/77/51.2%; T2 (65+): 1241/560/76/45.1%; T3 (reminder Yes): 3634/1721/182/47.4% — all match expected values exactly	PASS	CI error bars come from static tables and do not respond to slicers; bar values update correctly but CIs remain fixed.	Documented as a limitation: CI error bars are static and do not reflect filtered subsets. Noted for Week 8.	N/A	None (DA only)	outputs/dashboard/Power_BI_Refinement_and_Slicer_Steps.md; reports/test_evidence/dashboard_filter_expected_values.csv; screenshots T0–T3
Distance chart refinement (live dashboard)	Does the refined chart show the validated finding and keep correct values?	Apply colour rule, title and annotation; hover values; export	Values still 46.5 / 46.5 / 48.5 / 54.1%; only 15+ km highlighted; error bars intact	Values confirmed correct (46.5/46.5/48.5/54.1%). Annotation text present on the chart. Bar colour differentiation not applied to the live dashboard.	PASS (values) / NOT APPLIED (colour)	Colour differentiation between the three overlapping bands and the 15+ km band was not applied. All bars remain the same colour; may still imply a gradual trend to a casual reader.	Logged as a Week 8 limitation. Data and annotation are correct; visual encoding deferred to Week 8.	Carry into Week 8: apply conditional bar colour in Power BI and re-export.	None (DA only)	distance_chart_refinement_mockup.png; outputs/dashboard/Power_BI_Refinement_and_Slicer_Steps.md